

# Quantum Computing for Quantum Chemistry

## Module 4

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This lab focused on obtaining electronic energies using quantum phase estimation (QPE). Since the electronic Hamiltonian  $\hat{H}$  is Hermitian but not unitary, the unitary time evolution operator  $U(t) = \exp(-it\hat{H})$  can be simulated on a quantum computer. Applying  $U(t)$  to an energy eigenstate  $\Psi_j$  returns the same state multiplied by  $e^{-itE_j}$ , so QPE can be used to estimate a phase  $\theta$  such that  $e^{2\pi i\theta} = e^{-itE_j}$ , from which  $E_j$  can be recovered (once  $t$  and energy bounds are chosen). The QPE circuit was implemented with the most probable bitstring used to estimate  $\theta$ .

A simple one qubit phase gate example was used first to validate the implementation and confirm that exact  $n$  bit phases are recovered. QPE was then applied to  $H_2$  by preparing the state register in the unrestricted Hartree-Fock occupation state as a guess for the ground state. The time evolution circuit was implemented by a Trotterized approximation to  $U(t)$  from previous sections.

The measured phase was converted to an energy by choosing a time parameter  $t$  from a specified energy range and then selecting the integer  $m$  that placed the energy estimate inside that range.

Finally, the QPE ground state dissociation curve was compared against a full configuration interaction (FCI) reference curve. The comparison showed that FCI is variational in the chosen basis, but QPE is not, and can deviate due to finite precision, Trotterization error, and imperfect overlap between the guess state and the true eigenstate. Circuit depth and simulation time were also tracked as functions of  $n$ , showing the added cost of higher phase precision.